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In [4]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()
X = iris.data
y = iris.target

flowers = ['Rose', 'Aster', 'Orchids']
df = pd.DataFrame(X, columns = iris.feature_names)
df['species'] = [flowers[i] for i in y]

print(df.head())

sns.pairplot(df, hue = 'species')
plt.show()

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2)

model = KNeighborsClassifier(n_neighbors = 3)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

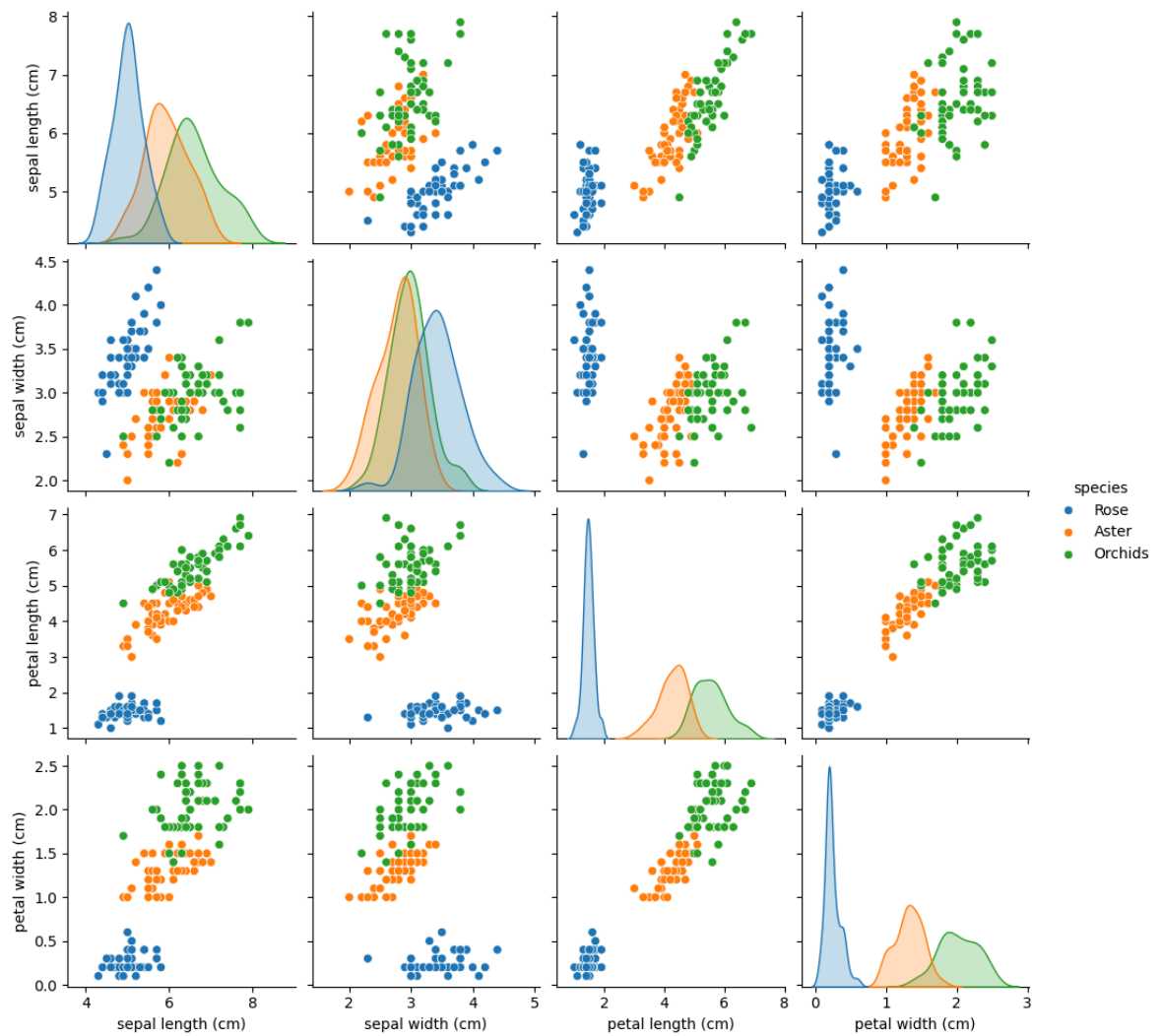
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot = True, cmap = "Blues")
plt.title("Confusion Matrix")
plt.show()

sample = [[5.1, 3.5, 1.4, 0.2]]
pred = model.predict(sample)
print("Predicted Flower:", flowers[pred[0]])

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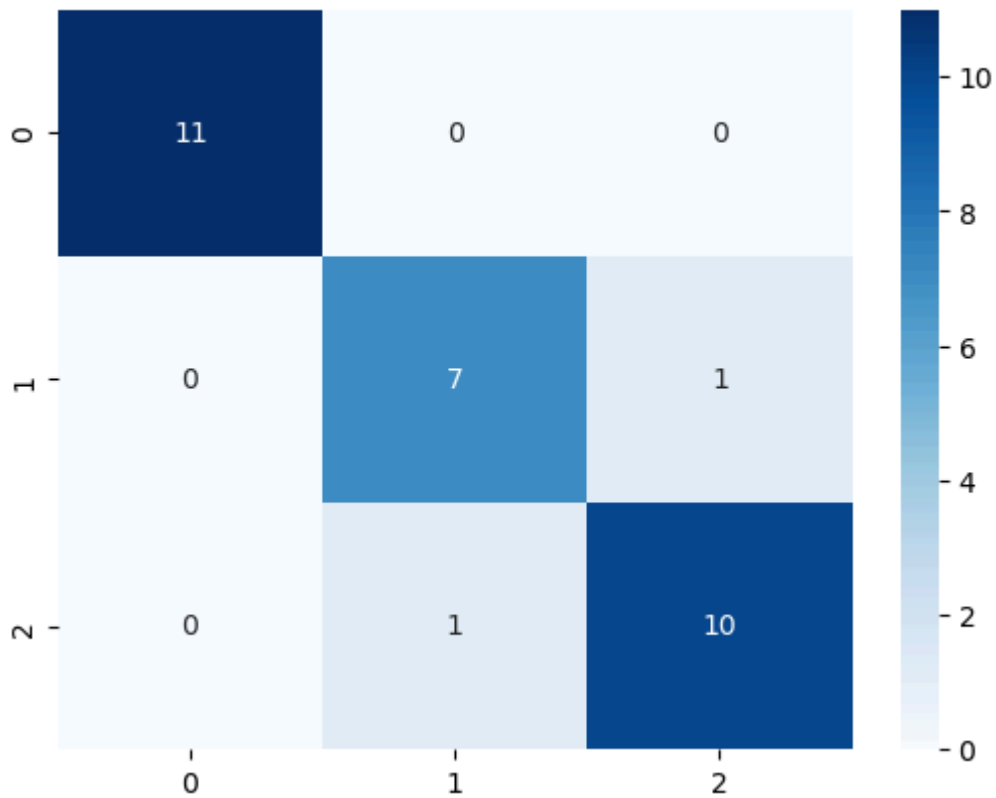
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species
0	Rose
1	Rose
2	Rose
3	Rose
4	Rose



Accuracy: 0.9333333333333333

Confusion Matrix



Predicted Flower: Rose

In []: SE-A,S3
Roll no-341